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PAPER

VME-EFD : A novel framework to eliminate the Electrooculogram artifact from single-channel EEGs

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Abstract

The diagnosis of neurological disorders often involves analyzing EEG data, which can be contaminated by artifacts from eye movements or blinking (EOG). To improve the accuracy of EEG-based analysis, we propose a novel framework, VME-EFD, which combines Variational Mode Extraction (VME) and Empirical Fourier Decomposition (EFD) for effective EOG artifact removal. In this approach, the EEG signal is first decomposed by VME into two segments: the desired EEG signal and the EOG artifact. The EOG component is further processed by EFD, where decomposition levels are analyzed based on energy and skewness. The level with the highest energy and skewness, corresponding to the artifact, is discarded, while the remaining levels are reintegrated with the desired EEG. Simulations on both synthetic and real EEG datasets demonstrate that VME-EFD outperforms existing methods, with lower RRMSE (0.1358 versus 0.1557, 0.1823, 0.2079, 0.2748), lower ΔPSD in the α band (0.10 ± 0.01 and 0.17 ± 0.04 versus 0.89 ± 0.91 and 0.22 ± 0.19, 1.32 ± 0.23 and 1.10 ± 0.07, 2.86 ± 1.30 and 1.19 ± 0.07, 3.96 ± 0.56 and 2.42 ± 2.48), and higher correlation coefficient (CC: 0.9732 versus 0.9695, 0.9514, 0.8994, 0.8730). The framework effectively removes EOG artifacts and preserves critical EEG features, particularly in the α band, making it highly suitable for brain-computer interface (BCI) applications.

1. Introduction

Utilizing portable systems greatly simplifies the process of recording electroencephalography (EEG) in both indoor and outdoor settings, particularly for applications involving brain-computer interfaces (BCI). For extended patient monitoring, prefrontal EEG recordings are predominantly favoured [1], and these find numerous applications [2–6]. Typically, recorded EEG data is susceptible to contamination from both physiological sources (such as cardiac and neural fluctuations) and non-physiological factors (like electrode pops, electrical shifts, and linear trend artifacts) [7, 8]. Unavoidable activities, such as eye movements or blinks, introduce electrooculogram (EOG) artifacts into the EEG signals, potentially leading to misleading interpretations of the brain activity within the range of 0 to 12Hz [9, 10].

The utilization of low-pass filters to eliminate EOG artifact has not yielded success due to the inadvertent loss of desired data alongside the artifacts [11, 12]. In addressing the removal of these artifacts, [13] discusses the implementation of regression and adaptive filters, with a reference signals. The Wiener filter is utilized for artifact removal without the need for a reference signal, but it encounters difficulties during the initial calibration phase [14]. Independent Component Analysis (ICA) and Canonical Correlation Analysis (CCA) [15–17] are employed to analyze artifact removal in multichannel signals. To utilize these techniques on single-channel signals, they are often coupled with other methods such as wavelet transforms (WT), Variational Mode Decomposition (VMD), Empirical Mode Decomposition (EMD), and Singular Spectrum Analysis (SSA), among others [18–25].

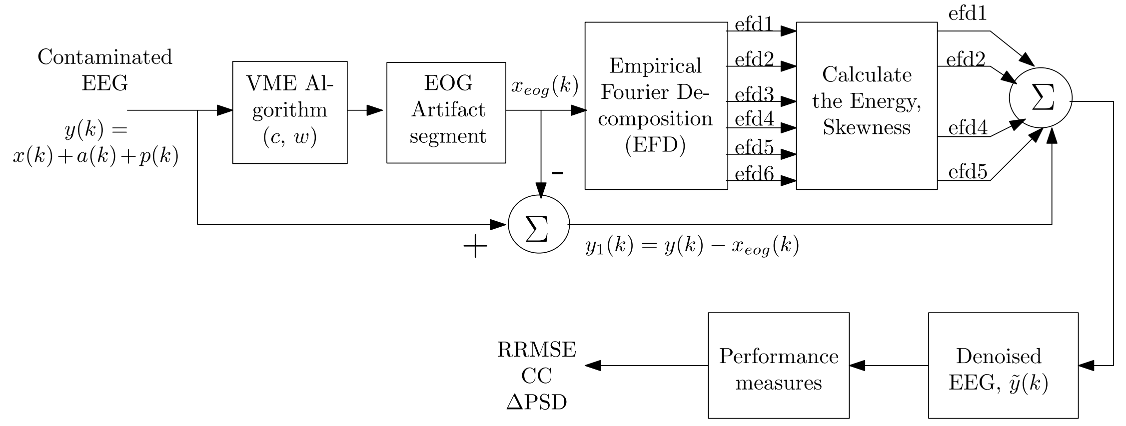

Figure 1. Proposed VME-EFD model implementation.

Recently, in [21], SSA is utilized to transform single-channel contaminated EEG signals into multivariate data. Here, SSA is repeatedly applied to divide the single-channel EEG into different components. At each stage, the signal is split into lower and higher frequency bands with the help of threshold value. The threshold value is represented as $fs/2^{n+1}$, n is number of stages, fs is sampling frequency. Subsequently, this multivariate data is inputted into ICA, which isolates the source signals into distinct independent components (ICs). Stationary Wavelet Transform (SWT) is employed on the artifact IC, facilitating thresholding to separate the actual artifact while preserving the EEG signal content. In [26], a method is introduced for eliminating EOG artifacts, combining overlap segmented adaptive singular spectrum analysis (Ov-ASSA) with adaptive noise canceler (ANC). The initial one or two reconstructed components from the Ov-ASSA technique are dynamically grouped and utilized as a reference EOG signal for ANC. Furthermore, in [27], the Variational Mode Extraction (VME)-DWT method is proposed for EOG artifact elimination. Here, the EOG segment is identified using the VME and then applies DWT to the identified segment, followed by skewness-based thresholding. In this, skewness values are utilized for identifying existence of eye blink. Based on the skewness value, the DWT decomposion will reach to maximum level of decomposition. In [29], the VME is combined to GMETV (Generalized Moreau Envelope Total Variation) filter to remove the EOG artifact. Both methods mentioned in [21, 27] involve the use of SWT/DWT with thresholding, and their performance relies on the selection of the mother wavelet and decomposition levels. However, it is important to note that thresholding introduces the risk of losing desired components due to the non-stationary nature of EEG signals, which can lead to a decrease in performance [25, 28].

To address the challenge of removing EOG artifacts, a novel framework known as VME-EFD has been proposed. This framework combines the VME (Variational Mode Extraction) [27] and EFD (Empirical Fourier decomposition) [30] algorithms. EFD [30], an efficient decomposition algorithm, primarily consists of two key steps. The first step is effective segmentation, based on the lowest minima criteria. The second step involves the zero-phase filter bank, designed according to the frequency segments obtained through effective segmentation. The process involves applying the interfered EEG to the VME technique, which separates it into two distinct segments: the desired EEG portion and the unwanted EOG portion. Subsequently, the extracted EOG segment is fed into the EFD algorithm, which decomposes it into multiple levels. For each decomposed level, parameters such as energy and skewness are calculated to identify the artifact. The decomposition level with the highest energy and skewness is identified as the artifact component and discarded. The remaining decomposition levels are then reintegrated with the desired EEG portion, resulting in a denoised EEG signal.

The document is organized as follows: section 2 elaborates on the proposed methodology and the existing methods being compared. Section 3 showcases the performance metrics and experimental results, and lastly, section 4 offers the paper's conclusion.

2. Methodology

2.1. Proposed VME-EFD

The depiction of the proposed VME-EFD model implementation can be observed in figure 1.

The VME algorithm is applied to the contaminated EEG signal, denoted as $y(k)$, to estimate the EOG artifact segment, $x_{eog}(k)$. Achieving an accurate estimation of $x_{eog}(k)$ necessitates careful selection of two crucial parameters: the compactness coefficient, denoted as c, and the center frequency, represented as ω_c. While previous work [31] suggested higher values for the compactness coefficient, this paper initiates the ‘c’ value at 15,000 and gradually decreases it in

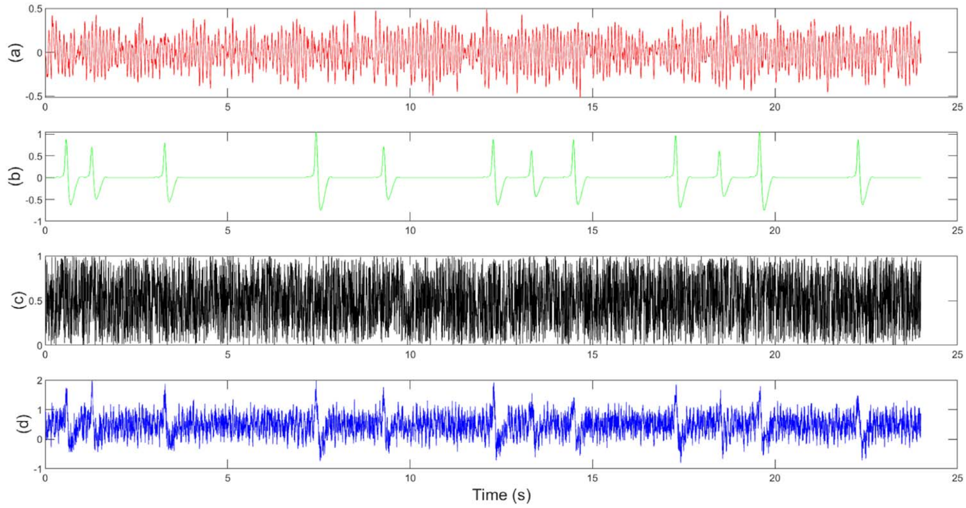

Figure 2. (a) Clean EEG, $x(k)$ (b) EOG , $a(k)$ (c) Random noise signal, $p(k)$ and (d) Interfered EEG signal, $y(k)$ with $-8dB$.

increments of 500 down to 1,000. Additionally, the center frequency ω_c is initialized at 12Hz, chosen because the EOG artifact primarily operates at lower frequencies, and it is then adjusted in increments of 0.5 Hz. The combination of c at 3500 and ω_c at 2.8 Hz yields a favourable estimation of the desired mode $x_{eog}(k)$ for the VME algorithm, leading us to adopt these specific parameter values.

The extracted EOG segment, $x_{eog}(k)$ is fed into the EFD [30] algorithm, which decomposes it into six levels as efd1 to efd6 as shown in figure 1. For each decomposed level, parameters such as energy and skewness are calculated to identify the artifact. The decomposition level, efd3 and efd6 has the highest energy and skewness and it is identified as the artifact component and discarded. The remaining decomposition levels are then reintegrated with the desired EEG portion, resulting in a denoised EEG signal, $\tilde{y}(k)$ as,

$$\tilde{y}(k) = y_1(k) + efd1 + efd2 + efd4 + efd5 \quad (1)$$

Where, $y_1(k) = y(k) - x_{eog}(k)$

2.2. Variational Mode Extraction (VME)

The VME, a refined adaptation of the VMD [24], serves to selectively extract precise modes from a given signals. While the VMD decomposes all possible modes, this can lead to increased computational complexity in certain applications, such as artifact removal, where many modes are unnecessary. Consequently, the VME emerges as a superior choice, particularly for tasks like artifact removal, due to its ability to efficiently target and extract the required modes while avoiding unnecessary computations.

The estimation of the wanted mode, $x_{eog}(k)$ is depending on the minimization criteria for bandwidth, which relies on the pre-defined center frequency ω_c, as proposed in [27, 31]. The procedure is outlined as follows:

(i) The input signal undergoes a Hilbert transform, given by $[(\delta(k) + j/k\pi)^* x_{eog}(k)]$.

(ii) To perform frequency shifting, the Hilbert transformed output signal is multiplied by $e^{-j\omega_c k}$, effectively shifting its frequency to ω_c: $[(\delta(k) + j/k\pi)^* x_{eog}(k)]^* e^{-j\omega_c k}$.

(iii) The bandwidth of the desired mode, $x_{eog}(k)$ is defined as the gradient of the squared second norm of the shifted input signal.

(iv) The goal is to minimize the bandwidth of $x_{eog}(k)$ which is expressed as:

$$||\partial_k[(\delta(k) + j/k\pi)^* x_{eog}(k)]e^{-j\omega_c k}||_2^2 \quad (2)$$

(v) The spectral separation of $x_{eog}(k)$ and $y_1(k)$ is obtained from a filter as:

$$f(\omega) = \frac{1}{\lambda(\omega - \omega_c)^2} \quad (3)$$

where λ is regulated coefficient.

To minimize the spectrum overlapping between the $x_{eog}(k)$ and $y_1(k)$ a penalty equation is applied:

$$J = ||F(k)^* y_1(k)||_2^2 \quad (4)$$

This approach aims to precisely extract the desired mode while minimizing interference from undesired modes by adjusting the bandwidth and employing a spectral filter.

2.3. Empirical Fourier Decomposition (EFD)

EFD, an efficient decomposition algorithm [30], primarily consists of two key steps. The first step is effective segmentation, based on the lowest minima criteria, where the input signal, $x_{eog}(k)$, is partitioned into multiple frequency segments, denoted as efd1, efd2, ..., efd6 (as illustrated in figure 1). This segmentation is determined by the magnitudes of the Fourier spectrum within each interval of [0, π], with the magnitudes sorted in descending order. The second step involves the zero-phase filter bank, designed according to the frequency segments obtained through

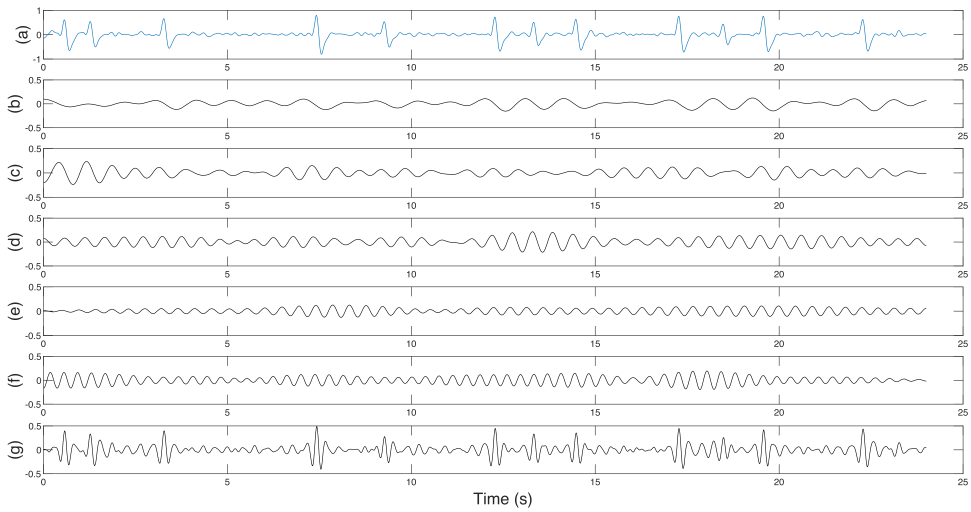

Figure 3. (a) Estimated the EOG artifact segment, $x_{eog}(k)$, (b) decomposed component efd1, (c) decomposed component efd2, (d) decomposed component efd3, (e) decomposed component efd4, (f) decomposed component efd5, and (g) decomposed component efd6.

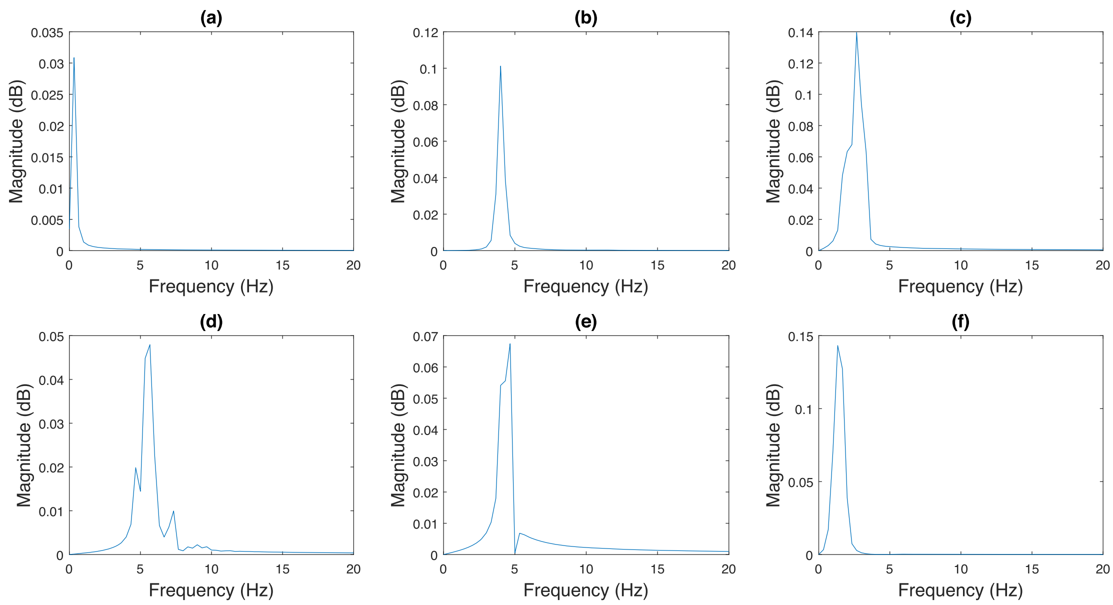

Figure 4. Power spectral densities (PSDs) of (a) efd1, (b) efd2, (c) efd3, (d) efd4, (e) efd5, and (f) efd6.

Table 1. Decomposed efds energy and skewness values.

	Energy	Skewness
efd1	0.2987	0.0008
efd2	3.0935	0.0001
efd3	13.0601	0.0319
efd4	3.4394	0.0058
efd5	3.8435	0.0073
efd6	11.7227	0.0350

effective segmentation. Within each of these frequency segments, a zero-phase filter functions as a band-pass filter, defined by its cut-off frequencies, $\omega_k - 1$ and ω_k, and exhibits no transitional phases. As a result, the zero-phase filter preserves the dominant Fourier spectrum component within the segment while excluding all other Fourier spectrum components located outside of the segment.

The EFD algorithm is summarised in the following steps.

Step 1: Begin by obtaining the Fourier spectrum of the signal to be decomposed, denoted as $x_{eog}(k)$, through the application of Fourier transform.

Step 2: Next, utilize the efficient segmentation, based on the Fourier spectrum obtained in Step 1, to determine the boundaries of the segments represented by ω_k.

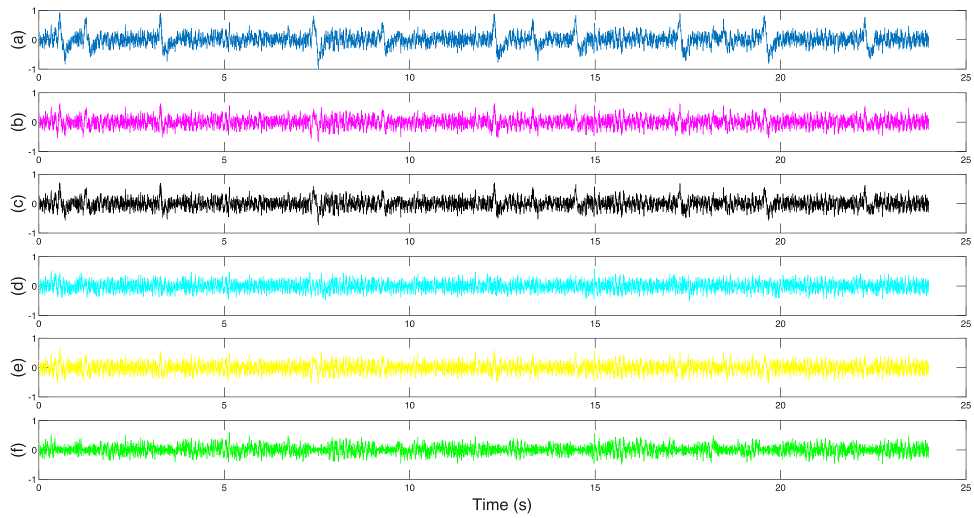

Figure 5. (a) Interfered simulated EEG, $y(k)$, Denoised EEG by (b) Ov-ASSA-ANC [26], (c) SSA-ICA-SWT [21], (d) VME-DWT [27], (e) VME-GMETV [29] and (f) Proposed VME-EFD.

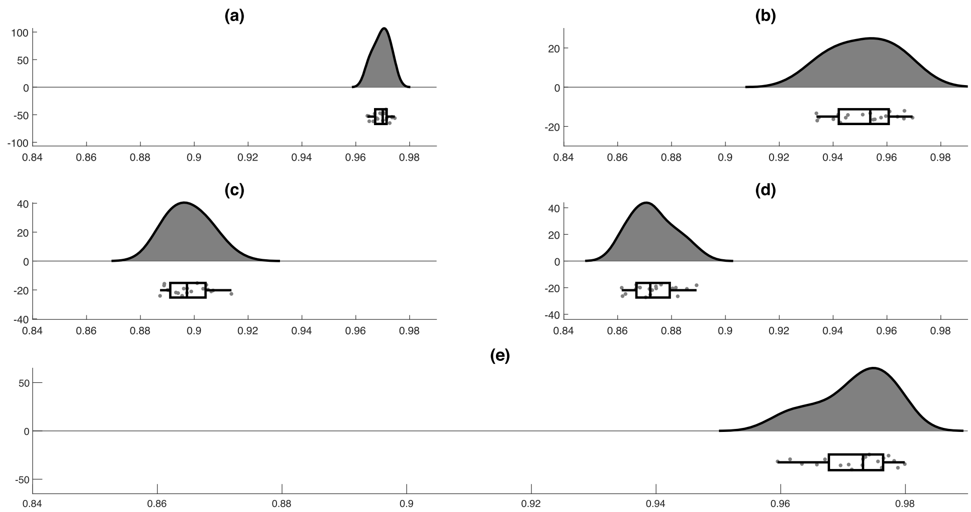

Figure 6. Raincloud plot representation of CC for (a) VME-GMETV [29], (b) VME-DWT [27], (c) SSA-ICA-SWT [21], (d) Ov-ASSA-ANC [26], and (e) the proposed VME-EFD method under good SNR conditions.

Step 3: Design a zero-phase filter bank denoted as $g(\omega)$, with the basis of ω_k obtained in Step 2.

Step 4: Proceed to obtain filtered signals $g_k(\omega)$ in the frequency domain, utilizing the filter bank $g(\omega)$ acquired in Step 3.

Step 5: Finally, obtain the decomposed components, denoted as $g_k(t)$, in the time-domain by employing inverse Fourier transforms on the frequency-domain signals $g_k(\omega)$ obtained in Step 4.

2.4. Existing techniques under comparison and Generation of the simulated data

2.4.1. SSA-ICA-SWT [21]

In the SSA-ICA-SWT approach [21], the single-channel EEG undergoes repeated SSA decomposition into multiple bands. In this [21], the authors adopts five decomposition levels and a window length of 50. Subsequently, ICA is employed to segregate hidden sources into independent components (IC) within these signal bands. Artifact ICs undergo soft thresholding using the SWT technique, which utilizes the db4 wavelet to compute detailed and approximation coefficients. These coefficients are then compared with a predefined soft threshold, where those exceeding it are set to zero, while the remaining coefficients are retained as the desired components.

2.4.2. Ov-ASSA-ANC [26]

Based on the Ov-ASSA-ANC method [26], the contaminated signal undergoes segmentation with a 5% overlap. Each segment is processed using adaptive SSA with a window length of 50. The first reconstructed component serves as the artifact, employed as a reference input for ANC to extract the denoised EEG.

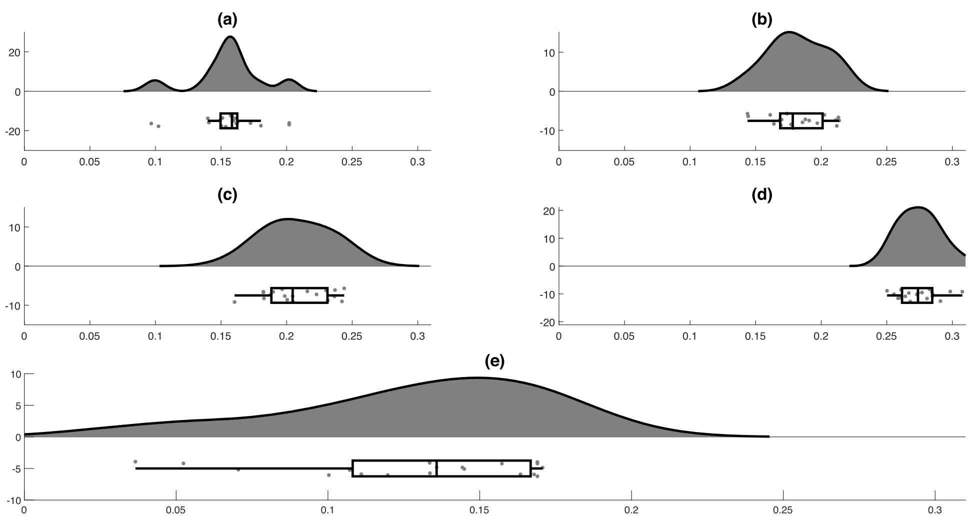

Figure 7. Raincloud plot representation of RRMSE for (a) VME-GMETV [29], (b) VME-DWT [27], (c) SSA-ICA-SWT [21], (d) Ov-ASSA-ANC [26] and (f) the proposed VME-EFD method under good SNR conditions.

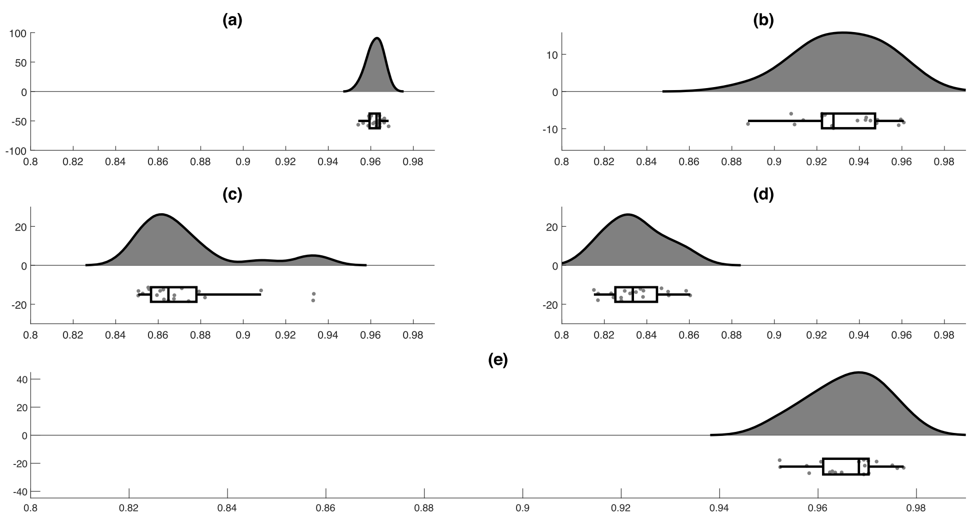

Figure 8. Raincloud plot representation of CC for (a) VME-GMETV [29], (b) VME-DWT [27], (c) SSA-ICA-SWT [21], (d) Ov-ASSA-ANC [26] and (f) the proposed VME-EFD method under mild SNR conditions.

2.4.3. VME-DWT [27]

In [27], the VME with compactness coefficient (c) and center frequency (ω) is set to 3000 and 3 Hz, respectively, and utilized to estimate the artifact component. Subsequently, the estimated artifact is applied to DWT followed by skewness-based thresholding.

2.4.4. VME-GMETV [29]

In [29], the VME technique is integrated with the GMETV filter to effectively eliminate the EOG artifact. The VME is responsible for extracting the artifact segment, which is then input into the GMETV filter. This filter discriminates between the desired EEG component and the artifact segment, effectively separating them. Subsequently, the EEG portion is reintegrated with the non-EOG signal, resulting in the production of the final denoised EEGs. Here, specific VME parameters are employed: a compactness coefficient (c) of 3500 and ω of 2.8 Hz, aimed at achieving accurate extraction of the wanted mode. Additionally, in the GMETV, a regularization parameter of 1.15 is utilized to enhance the estimation of EOG artifacts.

2.4.5. Generation of the simulated data

The generation of simulated data follows the equation (5).

$$y(k) = x(k) + a(k) + p(k) \tag{5}$$

Clean EEG data, $x(k)$ is acquired from [32], consisting of 54 records, each sampled at 200Hz, and an exemplar is displayed in figure 2 (a). The EOG artifact, $a(k)$ obtained from [33], includes artifacts of varying magnitudes and is positioned at different time intervals, as visualized in figure 2 (b). To maintain loyalty to real-time data, a random signal, $p(k)$ illustrated in figure 2 (c), is incorporated. The combination of these

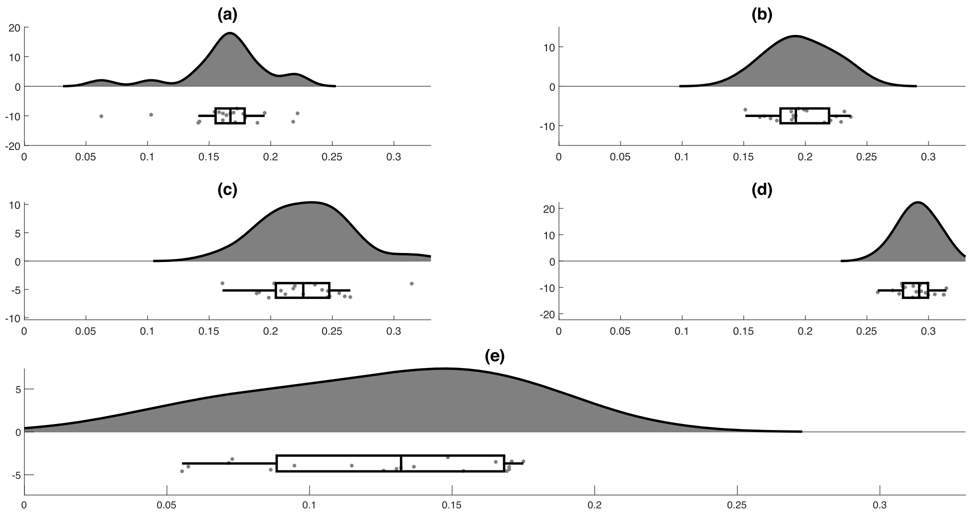

Figure 9. Raincloud plot representation of RRMSE for (a) VME-GMETV [29], (b) VME-DWT [27], (c) SSA-ICA-SWT [21], (d) Ov-ASSA-ANC [26] and (f) the proposed VME-EFD method under mild SNR conditions.

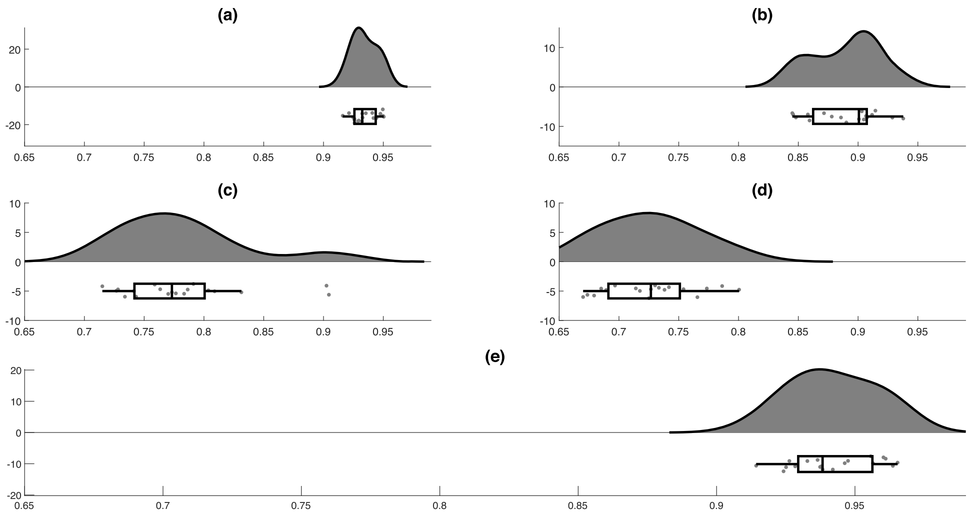

Figure 10. Raincloud plot representation of CC for (a) VME-GMETV [29], (b) VME-DWT [27], (c) SSA-ICA-SWT [21], (d) Ov-ASSA-ANC [26] and (f) the proposed VME-EFD method under poor SNR conditions.

three signals results in the construction of the interfered simulated data, $y(k)$ as depicted in figure 2(d).

3. Results

To assess the filtering capabilities of both existing and proposed methods, we examine two time domain parameters (RRMSE, CC) and one frequency domain (ΔPSD) parameter.

(i) The RRMSE quantifies the amplitude distortion present in the denoised EEG, which is represented as

$$RRMSE = \frac{RMS(x(k) - \tilde{y}(k))}{RMS(x(k))} \tag{6}$$

where, $x(k)$ represents the clean EEG and $\tilde{y}(k)$ denotes the filtered EEG.

(ii) The CC is expressed

$$CC = \frac{covariance(x(k), \tilde{y}(k))}{\sigma_{x(k)}\sigma_{\tilde{y}(k)}} \tag{7}$$

where σ indicates standard deviation

(iii) The ΔPSD (Difference in Power Spectral Density) is given

$$\Delta PSD = |PSD_{in} - PSD_{out}| \tag{8}$$

where, PSD_{in} and PSD_{out} are power spectrums of the contaminated and denoised EEGs.

3.1. Simulated EEG

The VME algorithm is applied to the contaminated EEG signal, $y(k)$, to estimate the EOG artifact segment,$x_{eog}(k)$, which is displayed in figure 3(a). The extracted EOG segment is given to the EFD algorithm [30], it will decompose the into six segments, (efd1,

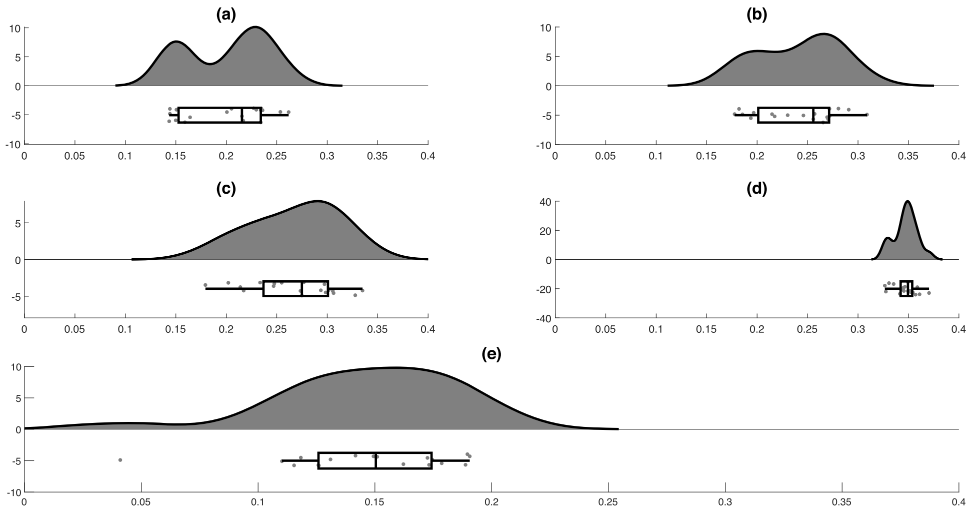

Figure 11. Raincloud plot representation of RRMSE for (a) VME-GMETV [29], (b) VME-DWT [27], (c) SSA-ICA-SWT [21], (d) Ov-ASSA-ANC [26] and (f) the proposed VME-EFD method under poor SNR conditions.

efd2, efd3, efd4, efd5 and efd6) as shown in figures 3 (b)-(g).

The power spectral densities (PSD) of these efd1 to efd6 are shown in figures 4 (a)–(f) respectively.

From the figure 4, it is noticed that, the efd with higher energy will reflect the EOG components. To estimate the exact artifact efds the parameters energy and skewness of each efd segment is calculated and displayed in table 1.

As per authors [7], the segment with higher energy and skewness is considered as EOG artifact segment, i.e., form the table 1, the edf3 and edf6 are identified as artifact segments and removed. The remaining decomposition levels, edf1, edf2, edf4 and edf 5are then reintegrated with the desired EEG portion, $y_1(k)$ resulting in a denoised EEG signal, $\tilde{y}(k)$, which is displayed in figure 5(f).

To assess the efficacy of both existing and proposed methods, we analyze a dataset comprising 19 records, each lasting 24 seconds. These records exhibit varying signal-to-noise ratios (SNRs) falling into three categories: poor SNR (below -8dB), mild SNR (between -8dB and 0dB), and good SNR (above 0dB). We evaluate the performance of both proposed and existing methods by computing their Root Relative Mean Square Error (RRMSE) and Correlation Coefficient (CC) values across these SNR categories. The results are organized into following raincloud [38] figures 6–12.

Based on figures 6–11, it is evident that the proposed method, VME-EFD outperforms existing methods [21, 26, 27, 29], achieving an average CC of 0.9732 versus 0.9695, 0.9514, 0.8994, 0.8730 and an average RRMSE of 0.1358 versus 0.1557, 0.1823, 0.2079, 0.2748 respectively under good SNR conditions.

As part of the statistical analysis, we conducted t-tests, ANOVA, and regression analyses to evaluate the efficacy of the proposed method. A paired t-test was conducted to compare the proposed method with each of the existing methods, using RRMSE and CC as performance measures. The results indicated that the calculated t-statistic (tcal) was significantly higher than the critical t-value from the t-distribution table. Consequently, the null hypothesis, which states that there is no difference between the mean RRMSE and CC values, was rejected. Furthermore, the two-tailed p-value (P_p) was less than 0.0001, indicating a highly significant result ($P_p < 0.01$). Therefore, it can be concluded that there is a statistically significant difference in RRMSE and CC between the existing and proposed methods.

Table 2 presents the ANOVA statistical analysis results used to assess the performance of the proposed algorithm. The analysis indicates that the proposed algorithm performs significantly better, with $P_p <$ 0.01.figure 12 presents the regression analysis of the proposed method. The normal probability plot in figure 12 shows that the residuals align closely with the diagonal line, indicating that the assumption of normality is satisfied. This alignment suggests that the model's residuals are approximately normally distributed, validating the regression analysis. Furthermore, the model demonstrates a high degree of accuracy, as indicated by an R^2 value of 94.8%.

3.2. Real EEG Data

The refinement and enhancement of the proposed method are facilitated through the use of simulated data testing. This approach involves iteratively adjusting parameters such as the number of decomposition levels in the EFD algorithm and the compactness coefficient, c, and center frequency, ω_c in the VME algorithm based on the performance results derived from simulated datasets.

To assess the effectiveness of this fine-tuned proposed method, it was evaluated using three real-time

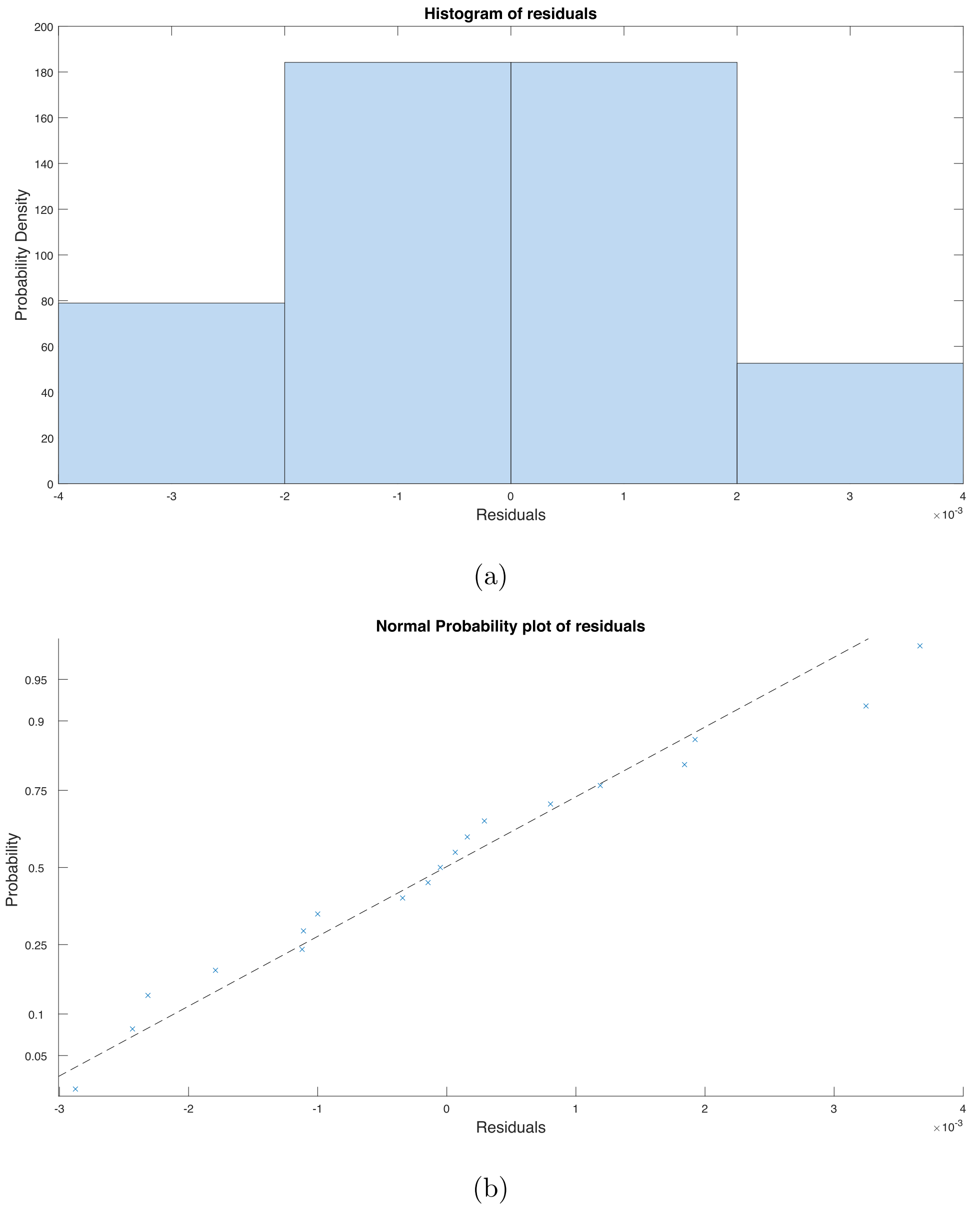

Figure 12. (a) Histogram of residuals, (b) Normal probability plot of residuals.

Table 2. ANOVA statistical analysis of CC and RRMSE between the reconstructed and original signals.

Performance Measures	Proposed versus [29] (P_p)	Proposed versus [27] (P_p)	Proposed versus [21] (P_p)	Proposed versus [26] (P_p)
CC	0.0065	5.7055e-08	6.26192e-29	1.42699e-32
RRMSE	0.0028	1.6255e-05	1.8250e-08	1.2471e-16

databases. Firstly, EEG data was sourced from the publicly available BCI-IV study database [34–36], comprising 64 records, each containing 23 channels. Here, the EEG activity was recorded using 22 Ag/AgCl electrodes, each positioned 3.5cm apart. The signals were acquired in a monopolar setup, with the left mastoid serving as the reference and the right mastoid as the ground. The data were sampled at a rate of 250 Hz and bandpass-filtered between 0.5 Hz and 100 Hz. The recording session consisted of three segments: (1) two minutes with eyes open, focusing on a fixation cross, (2) one minute with eyes closed, and (3) one minute involving eye movements. For experimental purposes, six segments (each consisting of 1800 samples) were selected, as depicted in figure 13.

The second database was obtained from the EEG BCI Dataset [37], which involved data collected from 55 participants performing both rapid serial visual presentation (RSVP) and P300 speller tasks. Here, the EEG records obtained by 32 Ag/AgCl active electrodes

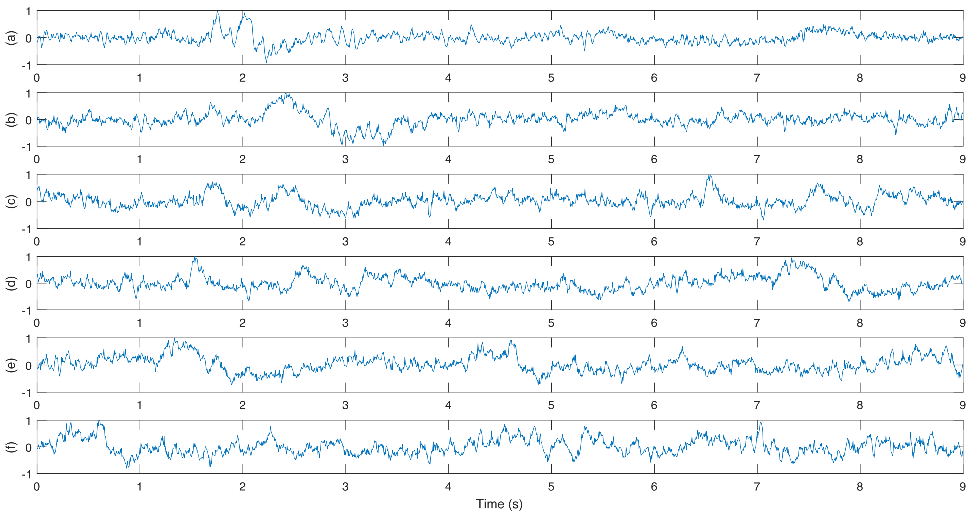

Figure 13. Real EEG data from BCI-IV (database-1) segments: (a) segment 1, (b) segment 2, (c) segment 3, (d) segment 4, (e) segment 5, and (f) segment 6.

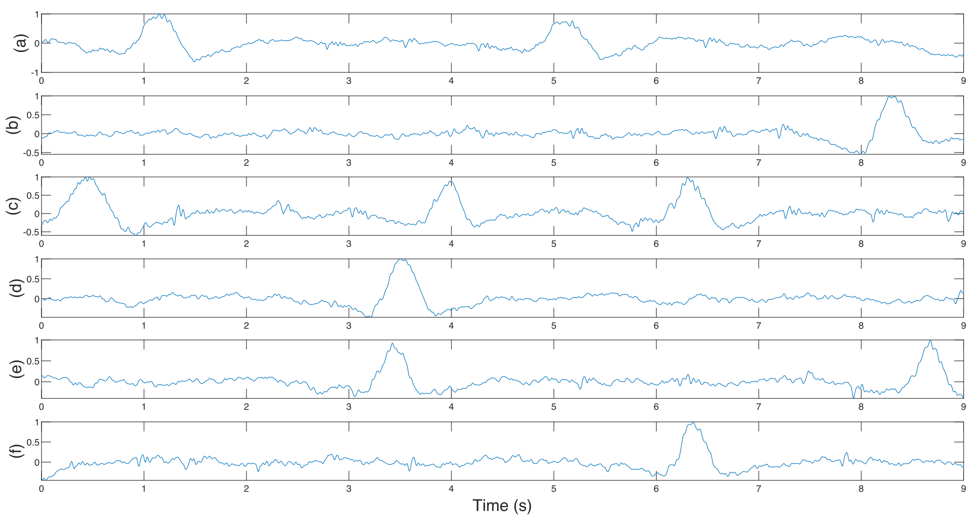

Figure 14. Real EEG data from the P300 Speller (database-2) segments: (a) segment 1, (b) segment 2, (c) segment 3, (d) segment 4, (e) segment 5, and (f) segment 6.

according to the international 10–20 system with sampling rate of 512Hz, which is down sampled to 250 Hz. This dataset comprises real data spanning a duration of 285 seconds. Within this dataset, the FP1 channel was identified as containing contaminated signals caused by EOG. Similar to the first dataset, six segments (each containing 1800 samples) were chosen for analysis, as illustrated in figure 14.

The third database was sourced from the EEG BCI2000 Dataset [39, 40]. This dataset comprises 64-channel EEG recordings from subjects who performed a series of motor and imagery tasks. The developers of the BCI2000 instrumentation system contributed these recordings to PhysioNet for brain-computer interface research. The EEG signals were captured using 64 electrodes following the international 10–10 system, excluding the electrodes Nz, F9, F10, FT9, FT10, A1, A2, TP9, TP10, P9, and P10, with a sampling rate of 160Hz. Similar to the first and second datasets, six segments (each containing 1440 samples) were chosen for analysis, as illustrated in figure 15.

Segment-1 from all the real-time Database-1, Database-2 and Database-3 is utilized in the existing and proposed models, and the resulting denoised signals are showcased in figures 16–18, respectively. Upon examination of figures 16–18, it becomes evident that the proposed VME-EFD model successfully eliminates artifact portions more effectively than the existing methods and retrieved the wanted EEG components. To conduct a numerical analysis of the filtering of both existing and proposed models on real databases, the performance metric ΔPSD is employed.

The average ΔPSD across all segments in database-1, database-2and database-3 for each frequency band is displayed in tables 3–5, respectively. The metrics provided in these tables serve as a valuable tool

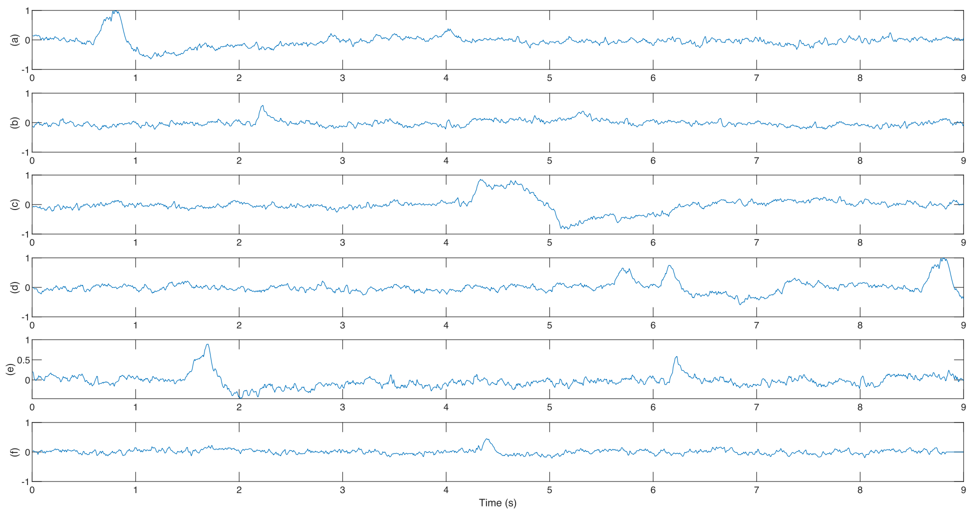

Figure 15. Real EEG data from the BCI2000 (database-3) segments: (a) segment 1, (b) segment 2, (c) segment 3, (d) segment 4, (e) segment 5, and (f) segment 6.

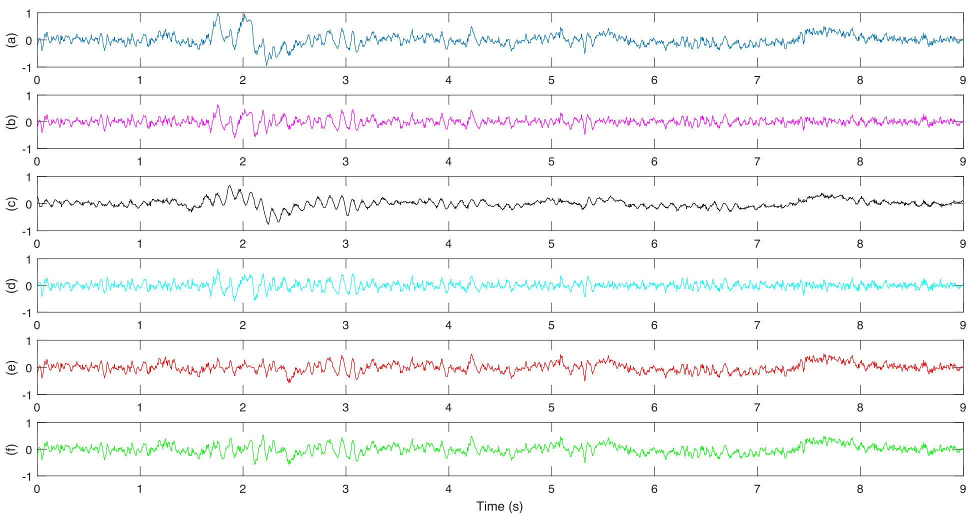

Figure 16. (a) Real EEG segment-1 (from database-1), Denoised EEG using (b) Ov-ASSA-ANC [26], (c) SSA-ICA-SWT [21], (d) VME-DWT [27] (e) VME-GMETV [29] and (f) the proposed VME-EFD method.

for assessing the performance of each algorithm in both identifying and quantifying the impact of ocular artifacts on the time domain.

The resulting denoised signals of Ov-ASSA-ANC method [26] are illustrated in figures 16(b), 17(b) and 18(b). Mean ΔPSD values across various frequency bands are computed and presented in the tables 3–5. Analysis of these tables reveals that the existing Ov-ASSA-ANC algorithm tends to eliminate EOG artifacts alongside desired EEG segments, notably in the α band with values of 3.96 ± 0.56, 2.42 ± 2.48 and 0.0061 ± 0.0044 respectively. A drawback of this approach [26] is its dependence on adjusting the overlapping portions for optimal performance. This limitation arises from the imperfect nature of the reference EOG input utilized for ANC.

In the SSA-ICA-SWT approach [21], the resulting denoised signals are depicted in figures 16(c), 17(c) and 18(c). Mean ΔPSD values across various frequency bands are computed and presented in the tables 3–5. Analysis of these tables reveals that the existing SSA-ICA-SWT method tends to eliminate EOG artifacts alongside desired EEG segments, particularly in the α band, with values of 2.86 ± 1.30, 1.19 ± 0.07 and 0.0022 ± 0.0014 respectively.

In VME-DWT method [27], the resulting denoised signals are depicted in figures 16(d),17(d) and 18(d). Mean ΔPSD values across various frequency bands are computed and displayed in the tables 3–5, revealing α band ΔPSD values of 1.32 ± 0.23, 1.10 ± 0.07 and 0.0012 ± 0.0018 respectively. Both methods mentioned in [21, 27] involve the use of SWT/DWT with thresholding, and their performance relies on the selection of the mother wavelet and decomposition levels. However, it is important to note that thresholding introduces the risk of losing

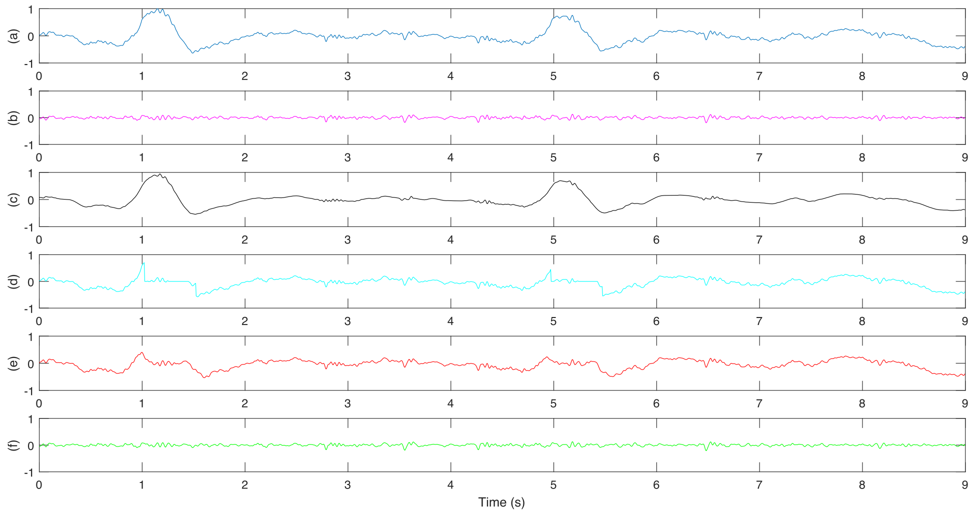

Figure 17. (a) Real EEG segment-1 (from database-2), Denoised EEG using (b) Ov-ASSA-ANC [26], (c) SSA-ICA-SWT [21], (d) VME-DWT [27] (e) VME-GMETV [29] and (f)the proposed VME-EFD method.

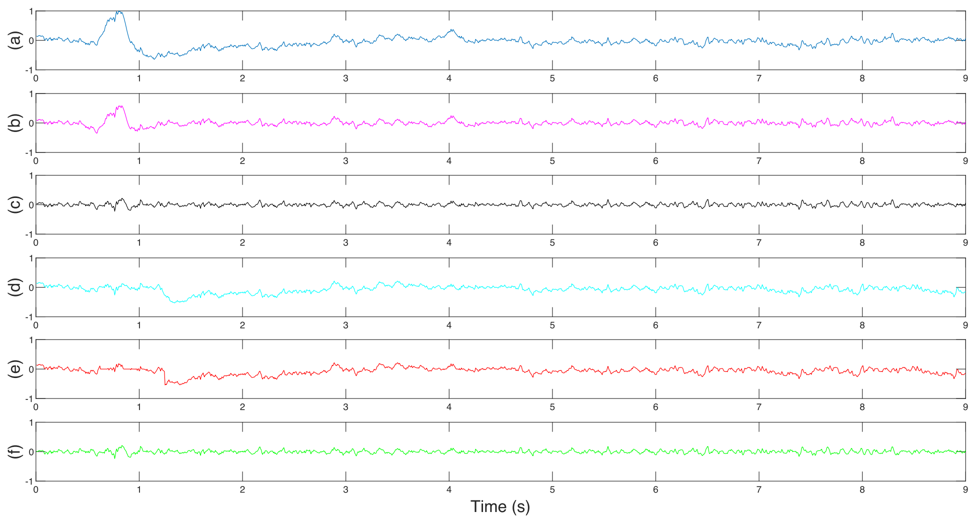

Figure 18. (a) Real EEG segment-1 (from database-3), Denoised EEG using (b) Ov-ASSA-ANC [26], (c) SSA-ICA-SWT [21], (d) VME-DWT [27] (e) VME-GMETV [29] and (f) the proposed VME-EFD method.

Table 3. Performance metric ΔPSD of real database-1.

	δ(0.5–4 Hz)	θ (4–8 Hz)	α(8–13 Hz)	β(13–30 Hz)	γ(30–100 Hz)
Ov-ASSA-ANC [26]	40.21 ± 0.92	1.52 ± 1.19	3.96 ± 0.56	2.62 ± 0.32	3.14 ± 1.15
SSA-ICA-SWT [21]	43.89 ± 3.10	9.67 ± 5.87	2.86 ± 1.30	5.65 ± 1.20	9.02 ± 1.24
VME-DWT [27]	45.35 ± 0.95	4.33 ± 0.52	1.32 ± 0.23	3.05 ± 0.05	1.054 ± 0.06
VME-GMETV [29]	46.27 ± 1.87	3.58 ± 1.47	0.89 ± 0.91	2.14 ± 2.67	2.86 ± 2.99
Proposed VME-EFD	46.02 ± 5.89	1.48 ± 1.18	0.10 ± 0.01	0.01 ± 0.01	0.05 ± 0.05

Table 4. Performance metric ΔPSD of real database-2.

	δ(0.5–4 Hz)	θ (4–8 Hz)	α(8–13 Hz)	β(13–30 Hz)	γ(30–100 Hz)
Ov-ASSA-ANC [26]	15.32 ± 5.27	6.87 ± 5.96	2.42 ± 2.48	0.17 ± 0.09	1.28 ± 0.29
SSA-ICA-SWT [21]	18.59 ± 1.04	4.33 ± 0.23	1.19 ± 0.07	3.10 ± 0.08	1.33 ± 0.28
VME-DWT [27]	17.38 ± 2.8	5.30 ± 1.65	1.10 ± 0.07	0.08 ± 0.02	1.13 ± 0.22
VME-GMETV [29]	19.84 ± 1.28	5.91 ± 2.80	0.22 ± 0.19	0.12 ± 0.02	1.01 ± 0.002
Proposed VME-EFD	23.76 ± 6.63	2.54 ± 0.22	0.17 ± 0.04	0.01 ± 0.02	0.91 ± 0.22

Table 5. Performance metric ΔPSD of real database-3.

	δ(0.5–4 Hz)	θ (4–8 Hz)	α(8–13 Hz)	β(13–30 Hz)	γ(30–100 Hz)
Ov-ASSA-ANC [26]	0.6237 ± 0.5032	0.0220 ± 0.0080	0.0061 ± 0.0044	0.0045 ± 0.0019	0.0021 ± 0.0014
SSA-ICA-SWT [21]	0.6338 ± 0.5115	0.0223 ± 0.0070	0.0022 ± 0.0014	0.0015 ± 7.5124e-04	3.4296e-04 ± 1.6845e-04
VME-DWT [27]	0.9070 ± 0.7224	0.0244 ± 0.0158	0.0012 ± 0.0018	5.2967e-04 ± 2.8655e-04	2.8199e-04 ± 1.6687e-04
VME-GMETV [29]	0.7534 ± 0.6555	0.0023 ± 5.6006e-04	4.8380e-04 ± 1.8071e-04	4.4969e-04 ± 1.8682e-04	2.3440e-04 ± 1.1380e-04
Proposed VME-EFD	1.0294 ± 0.6943	0.0155 ± 0.0072	5.7416e-04 ± 2.2954e-04	5.7190e-04 ± 3.4623e-04	2.7650e-04 ± 1.4979e-04

desired components due to the non-stationary nature of EEG signals, which can lead to a decrease in performance [25, 28].

In VME-GMETV method [29], the denoised signals are obtained and depicted in figures 16(e), 17(e) and 18(e) respectively. Mean ΔPSD values across various frequency bands are computed and presented in the tables 3, 4 and 5, showing α band ΔPSD values of 0.89 ± 0.91, 0.22 ± 0.19 and 4.8380e-04 ± 1.8071e-04 respectively. The primary limitation of this method lies in the adjustment of the regularization parameter in the GMETV filter.

In the proposed VME-EFD model, we consider specific VME experimental parameters: a compactness coefficient, c, set at 3500, and a center frequency, ω, at 2.8 Hz. These parameters play a crucial role in accurately estimating the unwanted EOG portion. Following this, the EOG segment extracted undergoes processing via the EFD algorithm, which decomposes it into six levels. At each decomposition level, metrics such as energy and skewness are computed to discern artifacts. The level exhibiting the highest energy and skewness is flagged as the artifact component and subsequently discarded. The remaining decomposition levels are then reintegrated with the desired EEG portion. This integration yields denoised EEG signals showcased in figures 16(f), 17(f) and 18(f) correspondingly. Additionally, mean ΔPSD values spanning various frequency bands are computed and summarized in the tables 3–5. Specifically, the α band ΔPSD values are recorded as 0.10 ± 0.01, 0.17 ± 0.04 and 5.7416e-04 ± 2.2954e-04 respectively.

Analysis of table 3, table 4 and table 5 reveals the efficacy of the proposed approach in eliminating EOG artifacts within the δ band while successfully recovering desired signals across other bands - θ, α, β, and γ - outperforming existing methods. Particularly noteworthy is the restoration of the α band component, vital for BCI applications. The proposed method demonstrates superior performance in this regard, as evidenced by the minimal ΔPSD values of 0.10 ± 0.01, 0.17 ± 0.04 and 5.7416e-04 ± 2.2954e-04 respectively, compared to alternative methods presented in the tables. Thus, affirming its suitability for BCI applications.

The proposed model offers several key advantages. The performance metrics show a 12.78% reduction in RRMSE and a 22.72% reduction in PSD in the α band, along with a 0.38% increase in CC, compared to the recent VME-GMETV [29] model. Additionally, the computation time for the proposed model is only 1.46 seconds, significantly less than the 6.24 seconds required by the VME-GMETV [29] model, indicating that the GMETV filter requires more computational resources.

4. Conclusion

This paper introduces a novel framework that integrates the VME method with the EFD technique to mitigate EOG artifacts. In this, the interfered EEG signal undergoes processing with the VME technique, which isolates the EOG segment. Subsequently, this segment is fed into the EFD algorithm, decomposing it into six levels. At each level of decomposition, various metrics including energy and skewness are computed to identify artifacts. The level with the highest energy and skewness is identified as the artifact component and discarded. The remaining decomposition levels are then merged with the desired EEG segment. Matlab simulations are conducted on simulated and real EEG databases to assess the performance of the proposed VME-EFD model in terms of RRMSE, CC, and ΔPSD. Experimental results demonstrate that the proposed technique surpasses existing methods, exhibiting lower RRMSE (0.1358 versus 0.1557, 0.1823, 0.2079, 0.2748), ΔPSD in the α band (0.10 ± 0.01 and 0.17 ± 0.04 versus 0.89 ± 0.91 and 0.22 ± 0.19, 1.32 ± 0.23 and 1.10 ± 0.07, 2.86 ± 1.30 and 1.19 ± 0.07, 3.96 ± 0.56 and 2.42 ± 2.48), and higher CC (0.9732 versus 0.9695, 0.9514, 0.8994, 0.8730). The performance metrics show a 12.78% reduction in RRMSE and a 0.38% increase in CC, compared to the recent VME-GMETV [29] model, 25.51% reduction in RRMSE and a 2.29% increase in CC, compared to the VME-DWT [27] model, 34.68% reduction in

RRMSE and a 8.21% increase in CC, compared to the SSA-ICA-SWT [21] model and 50.58% reduction in RRMSE and a 11.47% increase in CC, compared to the Ov-ASSA-ANC [26] model. Notably, the proposed method effectively removes EOG artifacts and accurately restores the α band component with minimal ΔPSD, making it suitable for BCI applications.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: The database links are as follows, Simulated data: https://data.mendeley.com/datasets/wb6yvr725d/1 Real time database 1: https://www.bbci.de/competition/iv/Real time database 2: https://doi.org/10.6084/m9.figshare.c.5769449.v1, real database 3: https://www.physionet.org/content/eegmmidb/1.0.0/S001/#files-panel.

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